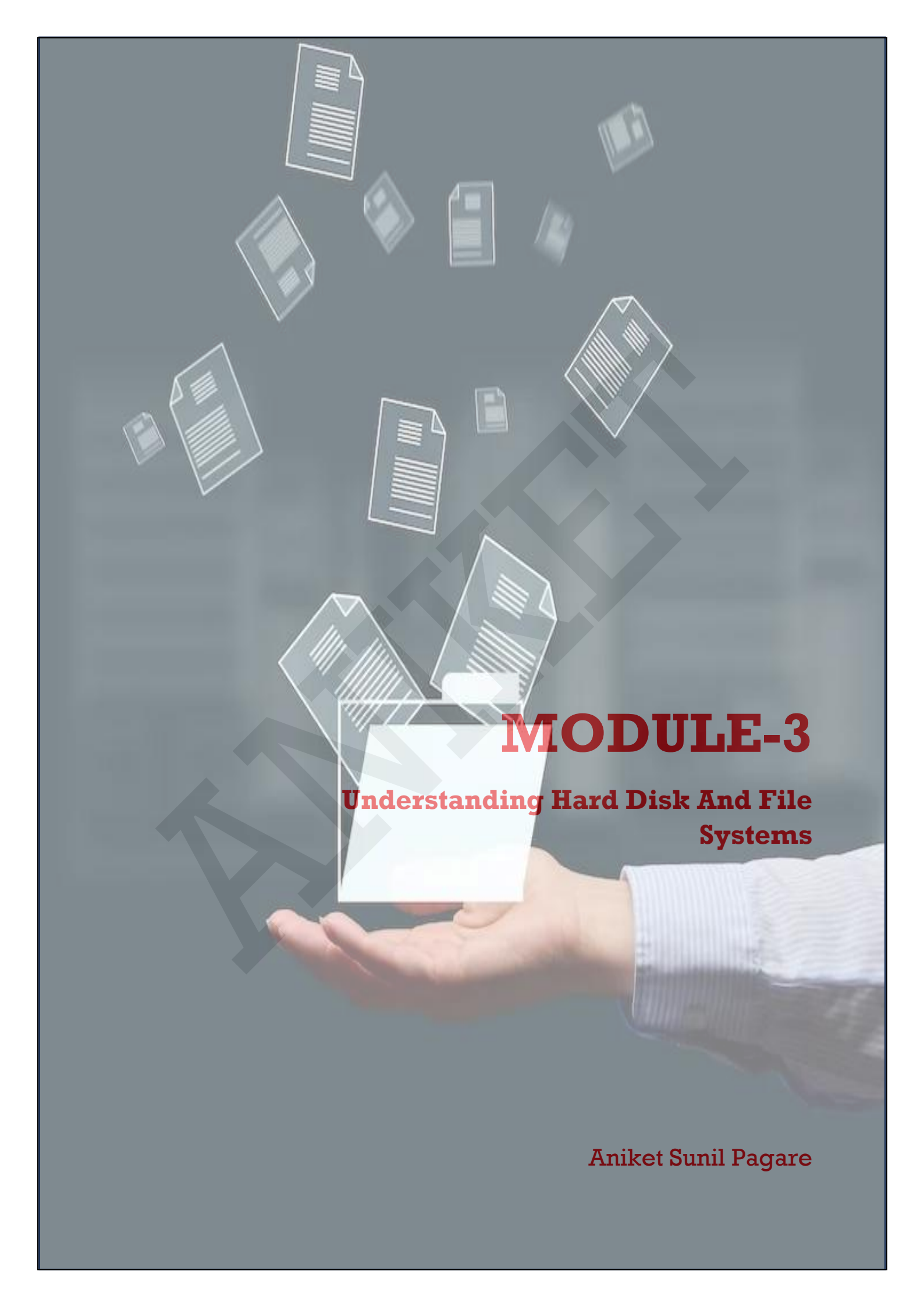


A hand in a light blue striped shirt sleeve holds a glowing white cube. Numerous document icons, some with text and others with images, float in the air around the cube. The background is a dark, textured grey.

MODULE-3

Understanding Hard Disk And File Systems

Aniket Sunil Pagare

Understanding Hard Disk And File System

◆ 1. What is a Hard Disk?

A **Hard Disk Drive (HDD)** is a non-volatile storage device that stores digital data using magnetic storage.

◆ Key Components:

- **Platters:** Magnetic disks that store data.
 - **Spindle:** Rotates the platters.
 - **Read/Write Head:** Moves across platters to read/write data.
 - **Actuator Arm:** Moves the head to the correct position.
 - **Controller:** Manages communication with the computer.
-

◆ 2. How Data is Stored on a Hard Disk

- Data is stored in **binary (0s and 1s)** on spinning platters.
 - Platters are divided into **tracks** and **sectors**.
 - **Sectors** are the smallest unit of storage (usually 512 bytes or 4 KB).
 - A group of sectors forms a **cluster** (allocation unit).
 - The operating system manages these structures using a **file system**.
-

◆ 3. Partitioning the Hard Disk

- A **partition** is a logical division of the physical disk.
- It allows separation of the OS, software, and data.

◆ Partition Types:

- **Primary Partition:** Can boot an OS.
- **Extended Partition:** Holds logical partitions.
- **Logical Partition:** Subdivision inside an extended partition.

◆ 4. File Systems Explained

A **File System** controls how data is stored and retrieved from a storage device.

◆ Common File Systems:

File System	Platform	Features
FAT32	Windows, USB	Simple, compatible, but max 4GB file size
NTFS	Windows	Supports permissions, encryption, large files
exFAT	Cross-platform	Good for large USB drives
EXT3/EXT4	Linux	Journaling support, large volume support
HFS+/APFS	macOS	Used in Apple devices

◆ File System Structures:

- **Boot Sector:** Contains system boot information.
- **File Allocation Table (FAT) / Master File Table (MFT):** Tracks file locations.
- **Root Directory:** The top-level directory.
- **Data Area:** Stores the actual file contents.

◆ 5. Deleted File Recovery Concepts

- When a file is deleted:
 - The **pointer** to the file is removed.
 - The actual data remains on disk until overwritten.
- Forensic tools (like FTK Imager, Autopsy) can recover **deleted files** by scanning unallocated space.

◆ 6. Slack Space and Unallocated Space

- **Slack Space:** Unused space at the end of a file's last cluster.
- **Unallocated Space:** Space not currently used by the file system but may still contain deleted data.

These areas often hold **valuable evidence** in forensic investigations.

◆ 7. Disk Imaging and Analysis

- A **forensic image** (bit-by-bit copy) includes all data:
 - Active files
 - Deleted files
 - Hidden data
 - Slack/unallocated space
 - Tools like **FTK Imager, EnCase, X-Ways** are used to capture and analyze this data.
-

🎯 Why Understanding Hard Disks and File Systems Matters in Forensics:

- Helps identify **where and how evidence is stored**
 - Enables recovery of **deleted or hidden data**
 - Ensures that evidence is acquired and analyzed **without altering the original**
 - Supports understanding of **file timestamps, metadata, and data carving**
-

✅ Key Learning Objectives:

1. **Understand the physical structure of hard disks**

- Learn how data is stored using platters, heads, and sectors.
- Grasp internal components like the actuator, spindle, and controller.

2. Identify and understand disk partitions

- Differentiate between primary, extended, and logical partitions.
- Understand partition tables (MBR vs GPT).

3. Learn the role of file systems

- Explore how file systems like FAT32, NTFS, EXT, HFS+, and APFS organize and manage data.
- Understand directory structures, metadata, and allocation units.

4. Understand how data is deleted and recovered

- Learn what happens when a file is deleted (logical deletion).
- Grasp the concept of recoverable data using forensic tools.

5. Explore key forensic concepts

- Slack space, unallocated space, and file carving.
- Volatile vs non-volatile data and how to access them.

6. Learn how to create and analyze disk images

- Importance of imaging over direct access to preserve evidence integrity.
- Use of forensic tools like FTK Imager for disk acquisition.

7. Apply knowledge to forensic scenarios

- How to interpret storage layout during an investigation.
- Finding hidden, deleted, or intentionally obfuscated data.

Practical Outcome:

After completing this module, you should be able to:

- Explain how hard drives work and how data is stored.
 - Identify partitions and file system structures.
 - Use forensic tools to create disk images.
 - Recover deleted files using file system knowledge.
 - Extract and analyze evidence from slack and unallocated space.
-

Lab-1 Analyze file system of a linux Image

Analyzing the file system of a Linux disk image is a **critical step in digital forensics**. A disk image is a **bit-by-bit copy** of a storage medium (such as a hard disk, SSD, or USB), and it often contains evidence that is crucial to an investigation.

This process involves using specialized tools and techniques to:

- Understand the file system structure,
- Locate and extract files (including deleted or hidden ones),
- Identify evidence such as user activities, malware traces, and access logs.

A) Analyze File System Using Autopsy

Autopsy is a **free, open-source digital forensics platform** used to investigate **hard drives, disk images, smartphones, and memory dumps**. It provides a **Graphical User Interface (GUI)** for the **Sleuth Kit** (a powerful command-line forensic tool), making forensic analysis easier and more visual for investigators.

Objectives of Autopsy -:

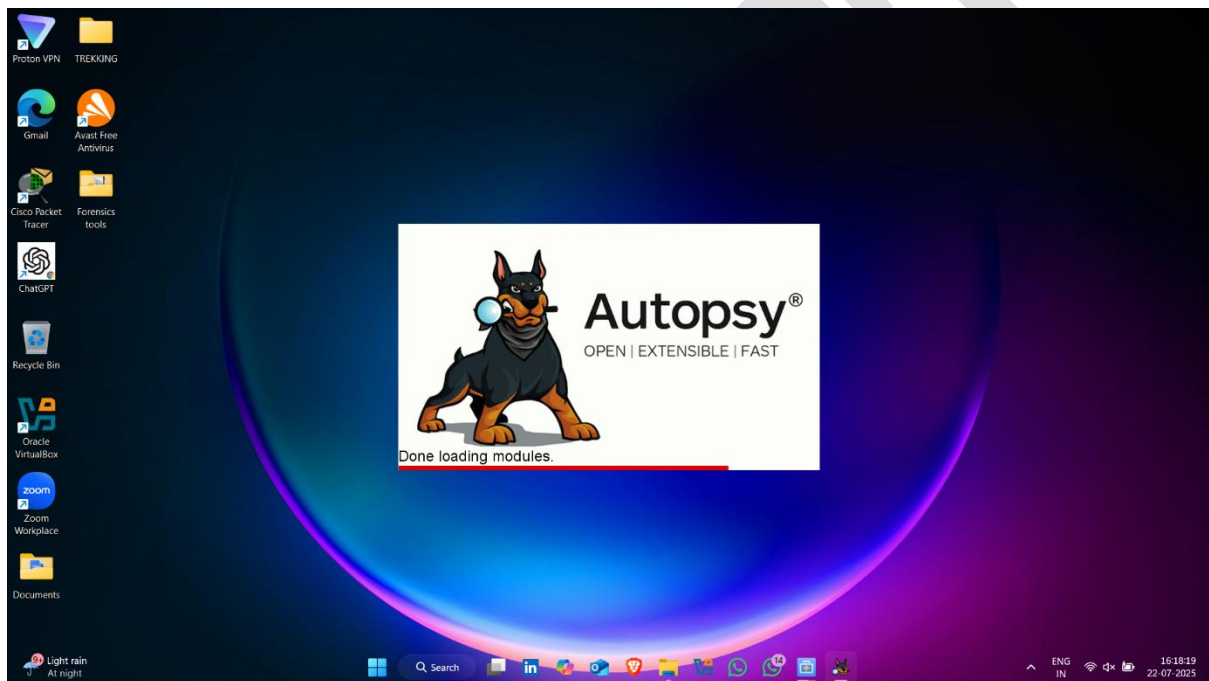
1. To provide a GUI-based platform for forensic analysis using The Sleuth Kit.
2. To recover and examine deleted or hidden files from digital storage.
3. To extract artifacts such as browser history, emails, and chat logs.
4. To generate a timeline of user activities and file events.
5. To support keyword and hash-based file searching for evidence identification.
6. To automate and streamline digital investigations for faster [results](#).

7. To create court-admissible forensic reports with detailed findings.
8. To allow modular integration of plugins for extended forensic capabilities.

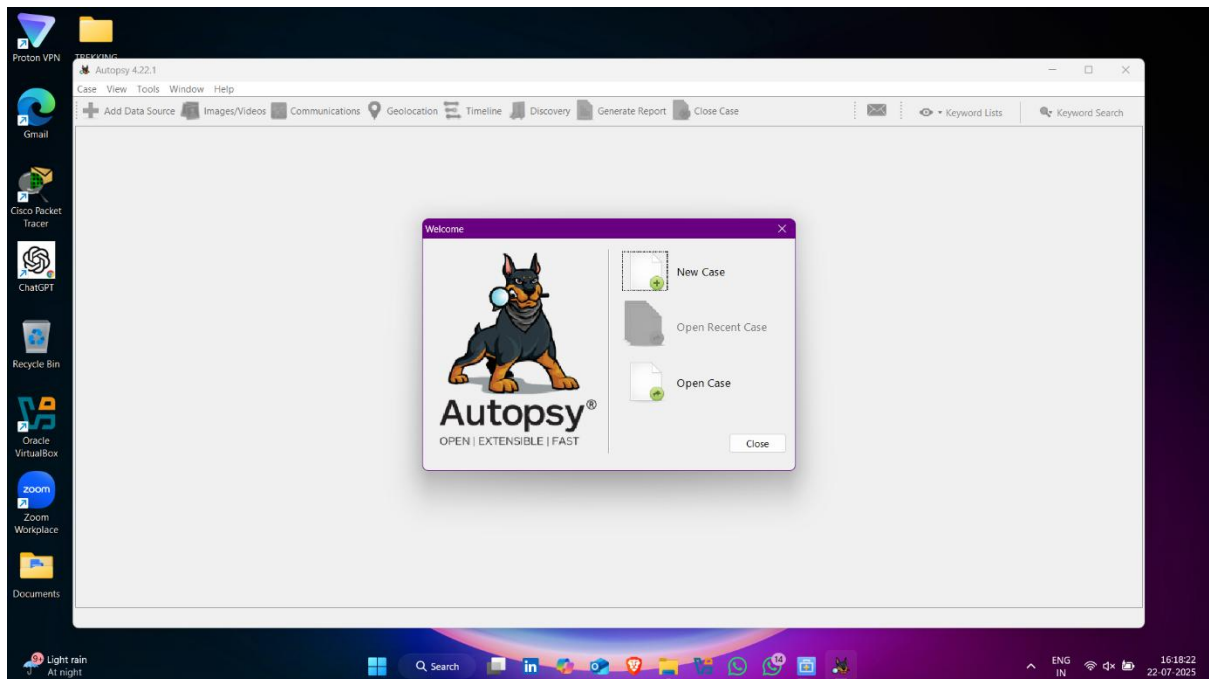
Download Link : <https://www.autopsy.com/download/>

How to use it -:

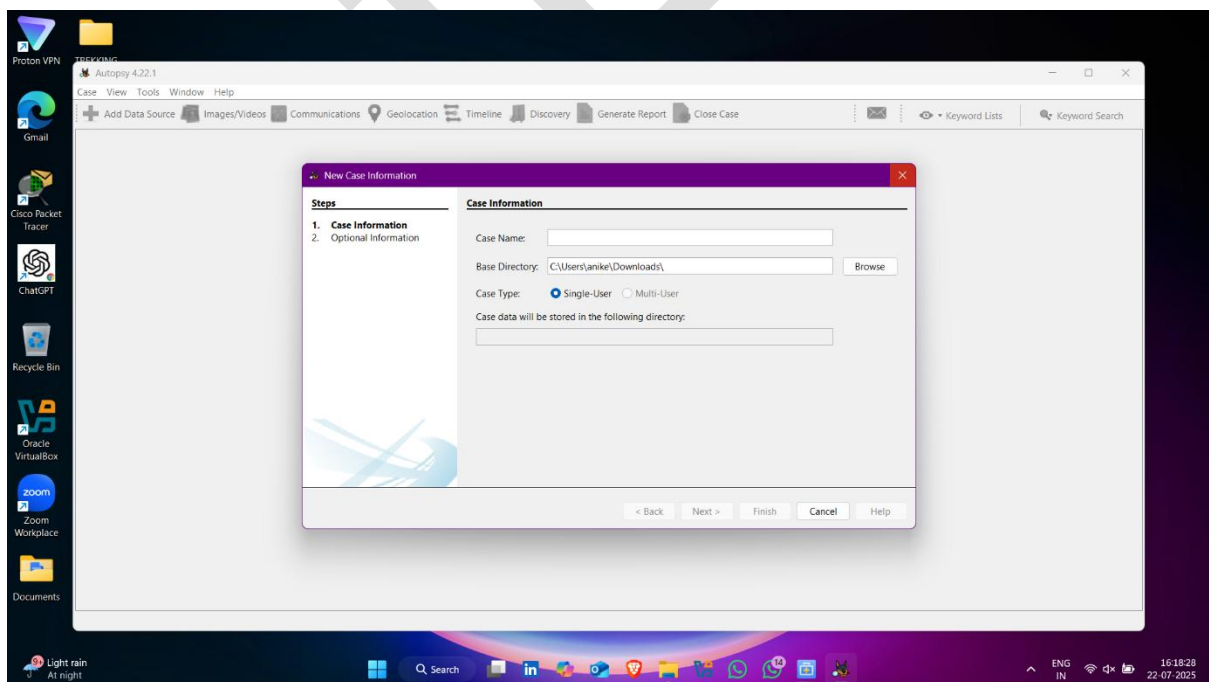
- Open Autopsy

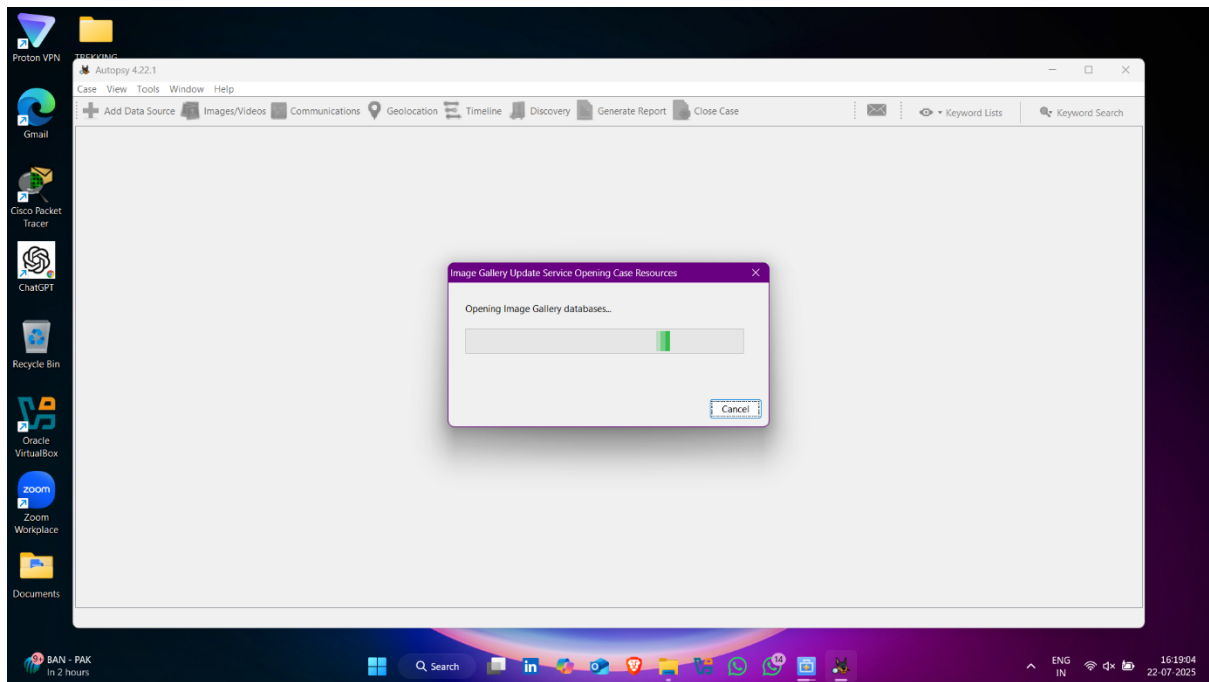


- Autopsy **welcome** windows will appear along with Autopsy main window in the background , In **Welcome** Window, click on **New Case**

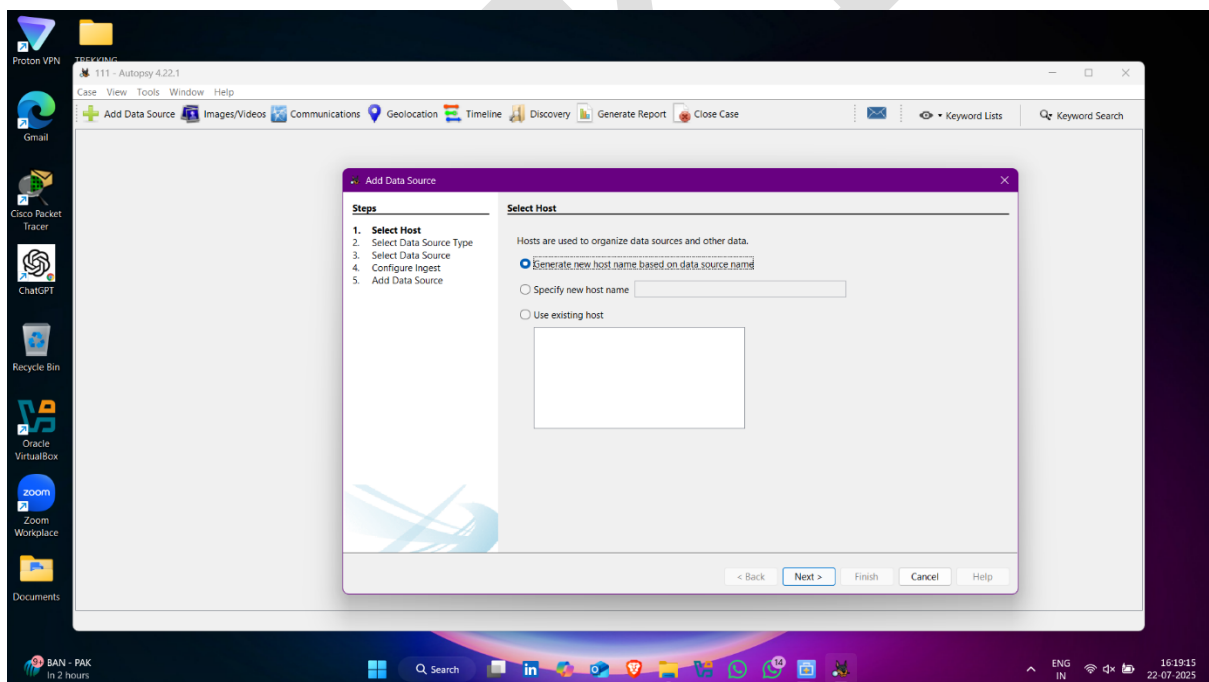


- Enter case name and base directory location

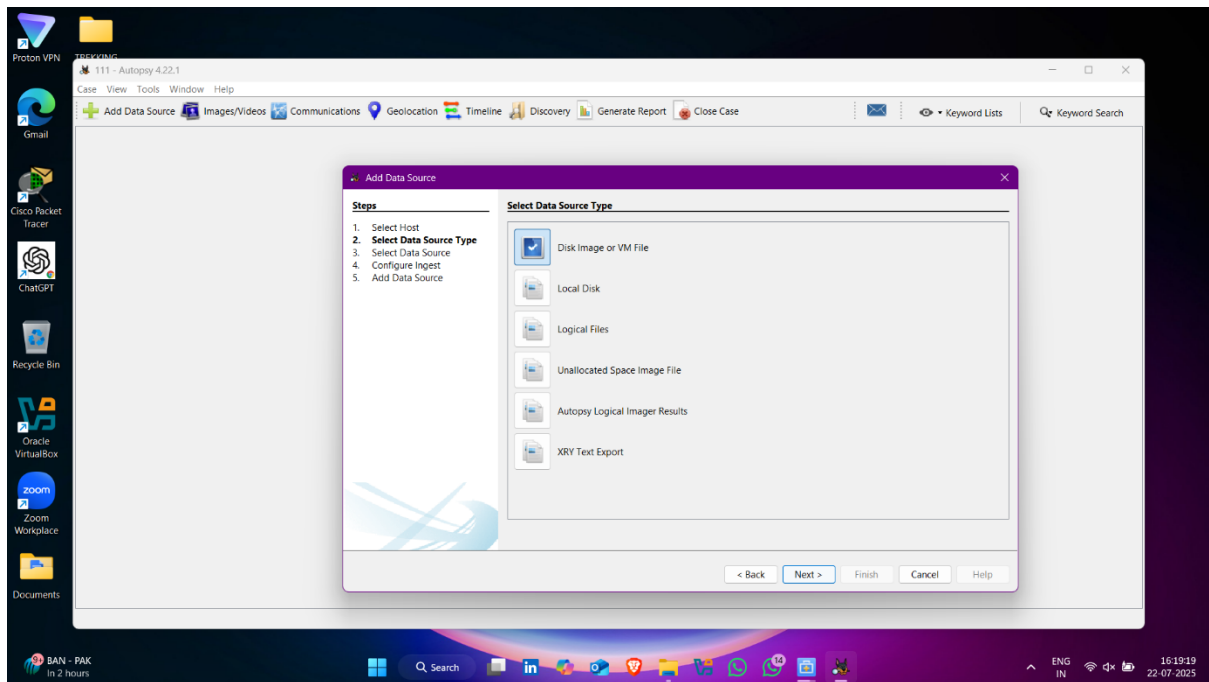




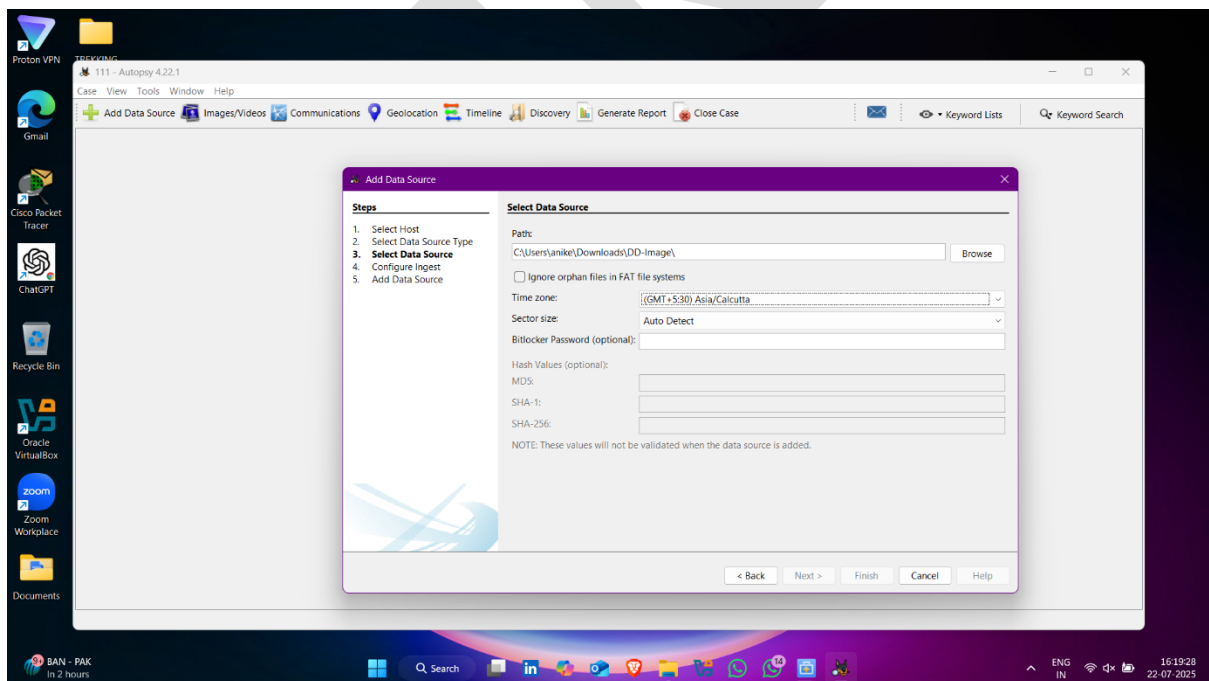
- Click on first option and then click on **next**



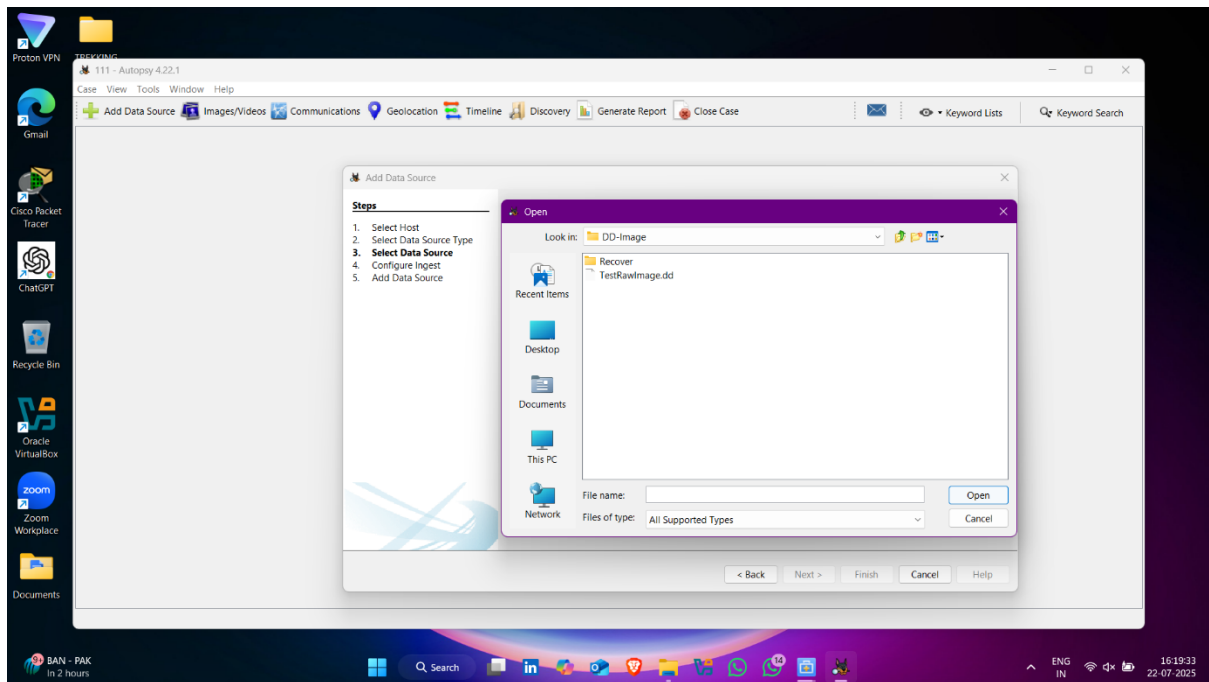
- Select first option – **Disk Image or VM File**



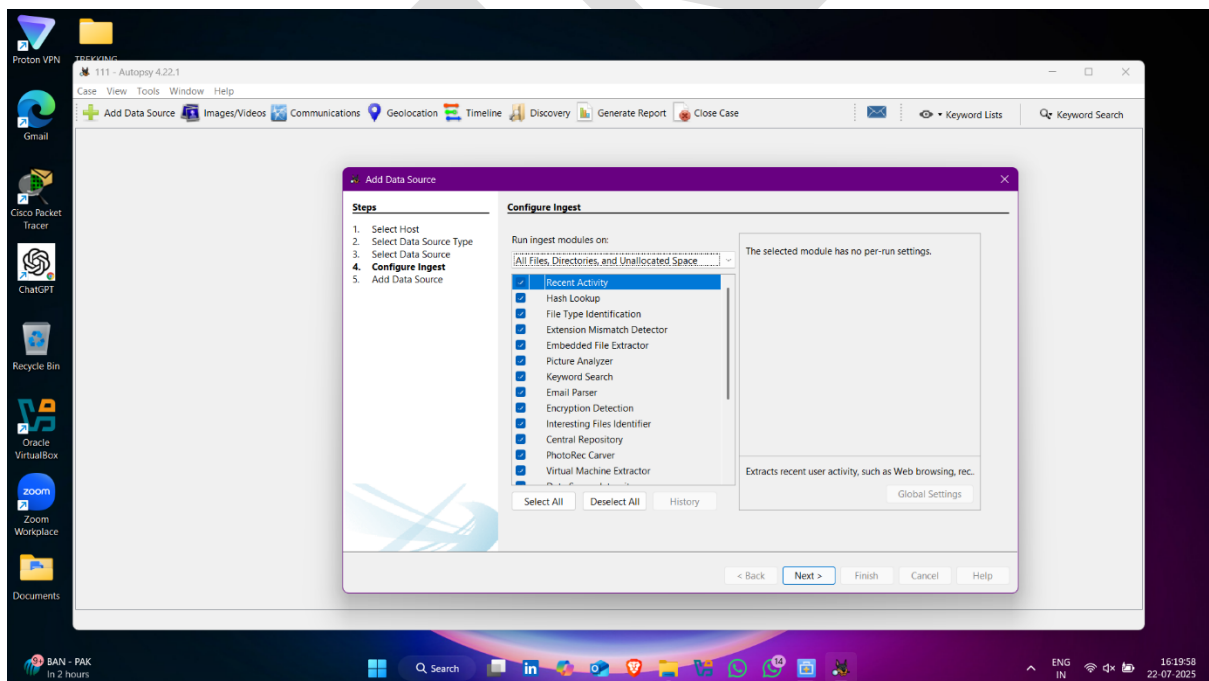
- Click on Browse Option



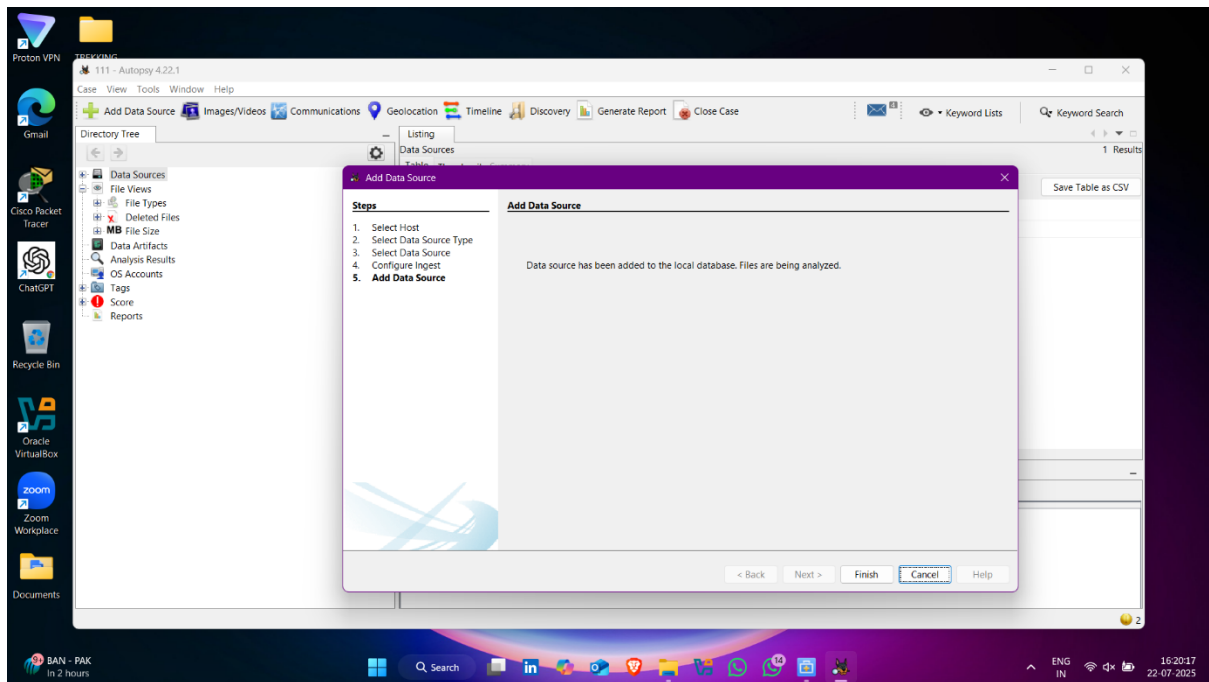
- **Select Image File**



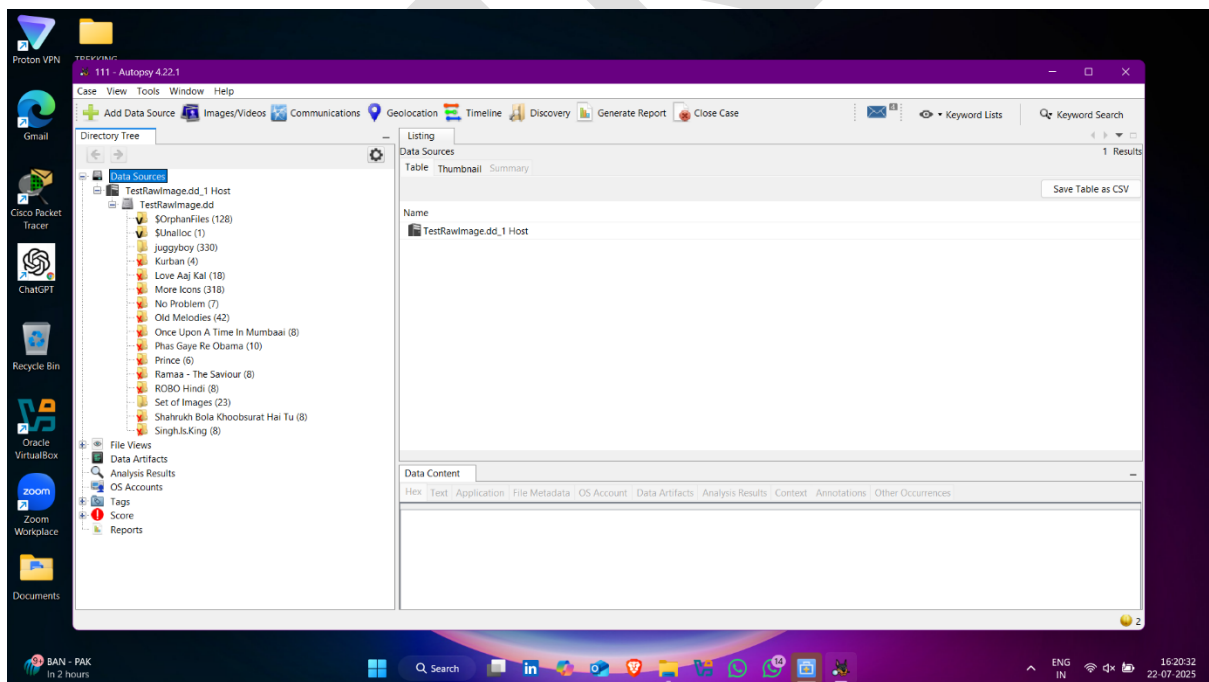
- **Select all and then click on Next**

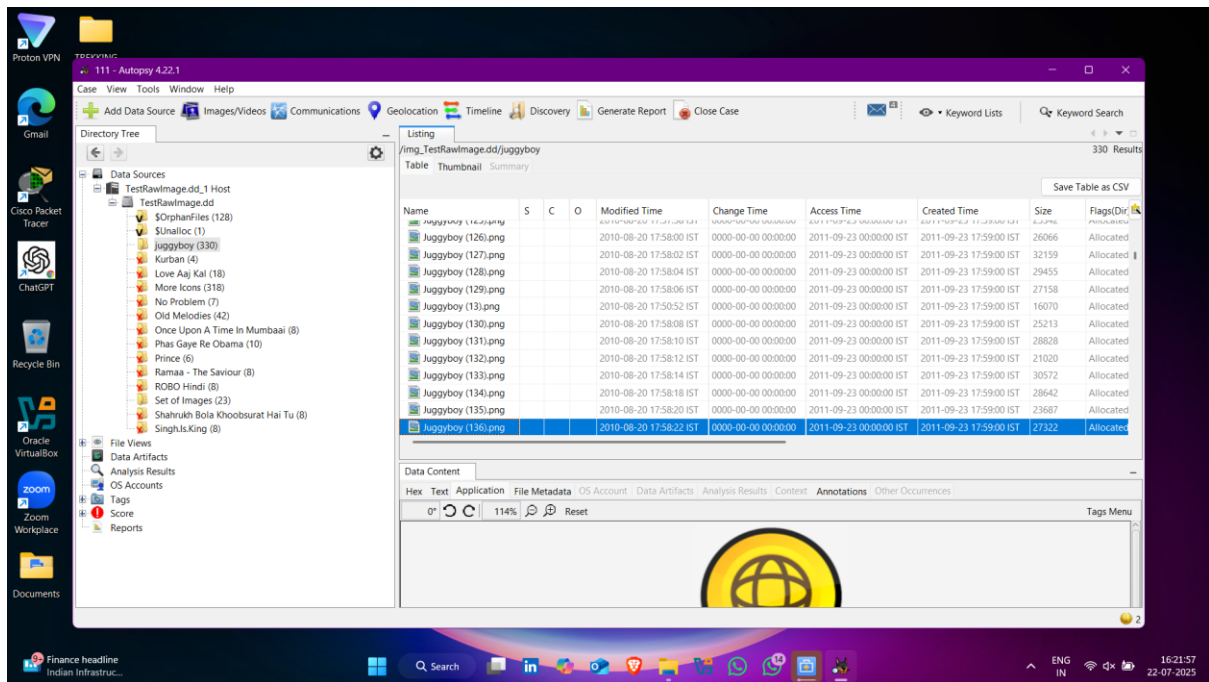


- Click on **Finish**



- Image data recover 🖱️ ✅





Lab-3 -: Recover Deleted Files From Hard Disk

In **computer forensics**, recovering deleted files from a hard disk is a critical task used to retrieve digital evidence that may have been intentionally or accidentally deleted. This process involves identifying and extracting files that are no longer visible through the standard file system interface but still exist in the **unallocated space** or **disk slack**.

When a file is deleted, it is **not immediately erased** from the disk. Instead, the operating system marks the space as "available" for reuse, and the actual data remains intact until it is overwritten by new data. Forensic tools can scan the disk for such data remnants and recover them by analyzing file headers, structures, and metadata.

To maintain evidence integrity, professionals never work directly on the original hard drive. Instead, they create a **bit-by-bit forensic image** of the drive using tools like **FTK Imager** or the `dd` command. Recovery is then performed on the image using tools such as **WinHex**, **Autopsy**, **PhotoRec**, or **The Sleuth Kit (TSK)**, which allow investigators to locate deleted files through file carving or MFT (Master File Table) analysis.

This process not only recovers deleted content but also helps in **timeline reconstruction**, **user activity tracking**, and **evidence correlation**. It plays a crucial role in solving cybercrimes, fraud investigations, and legal proceedings where deleted data could be key evidence.

A) Recover Deleted Files Using WinHex Tool

WinHex is a professional hexadecimal and disk editor widely used in **computer forensics**, **data recovery**, and **low-level data analysis**. It enables forensic experts to examine and recover deleted data directly from hard drives, memory cards, disk images, and other storage devices by working at the **binary level**.

When files are deleted from a system, they are not immediately erased but rather marked as available space by the file system. WinHex helps investigators locate this residual data in the **unallocated space** by parsing file systems (like NTFS, FAT, or EXT),

analyzing metadata (e.g., Master File Table in NTFS), and identifying file signatures.

The tool allows forensic investigators to:

- View and edit raw disk sectors.
- Identify and recover deleted files using file carving techniques.
- Create and analyze disk images.
- Preserve evidence integrity by working on a bit-by-bit image instead of the original drive.

WinHex's ability to work at the **lowest level of the storage media**, its support for **file system interpretation**, and its **data recovery functions** make it essential for uncovering deleted or hidden evidence in digital investigations.

Objectives of WinHex Tool :

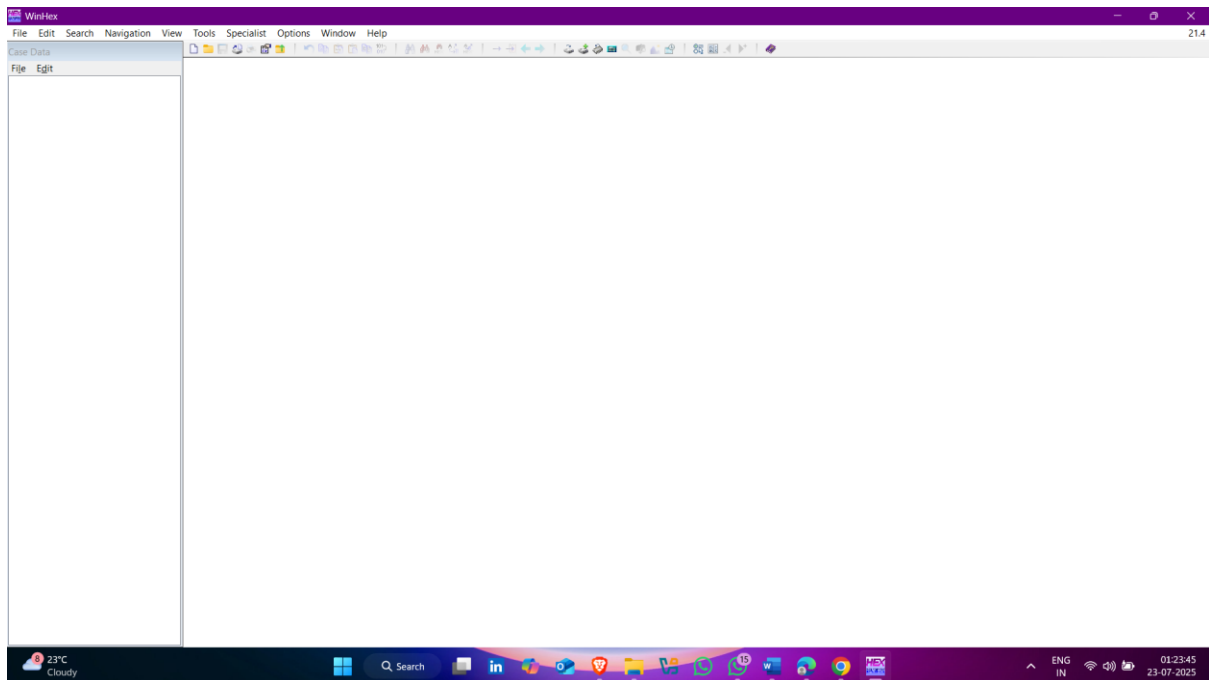
1. To recover deleted or lost files from digital storage.
 2. To analyze and edit raw disk sectors at the hexadecimal level.
 3. To examine and interpret file systems like NTFS, FAT, and EXT.
 4. To carve and extract files based on their headers and footers.
 5. To create, clone, and verify forensic disk images.
 6. To investigate hidden or slack space for concealed data.
 7. To support keyword and pattern searching in binary data.
 8. To maintain data integrity by working on read-only disk images.
 9. To assist in timeline and metadata analysis for forensic reporting.
 10. To provide a powerful environment for manual forensic investigation.
-

Download Link :

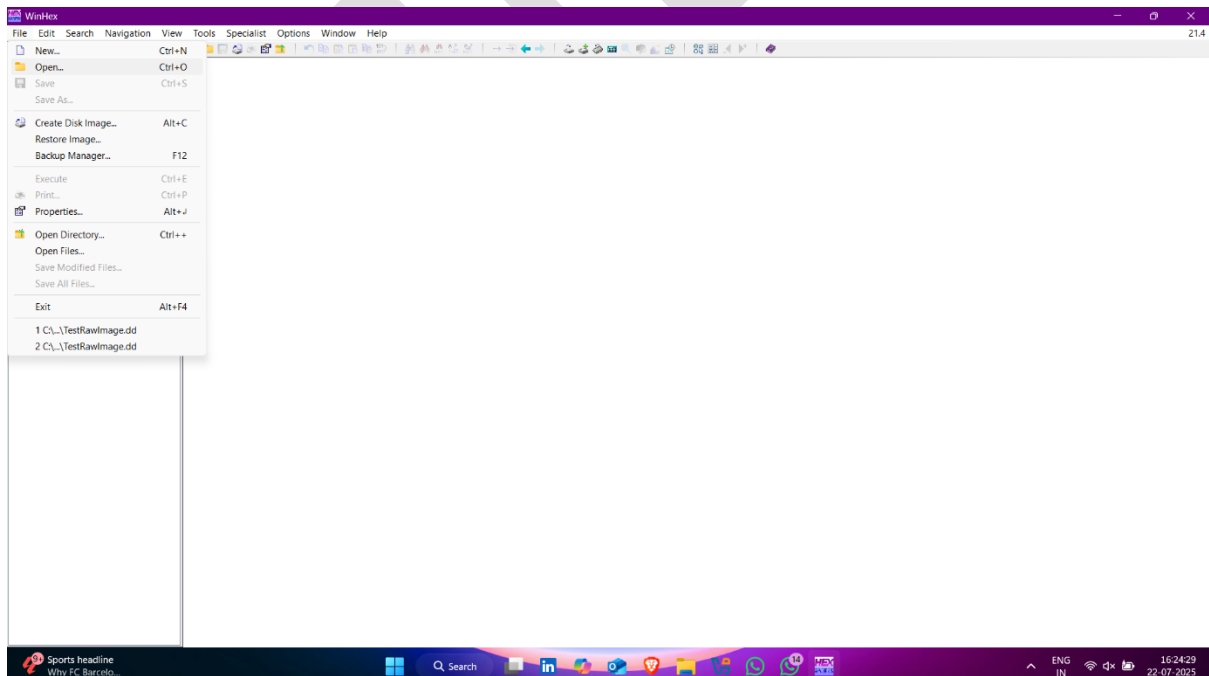
<https://winhex.en.softonic.com/download?installerType=riseInstaller>

How to use it -:

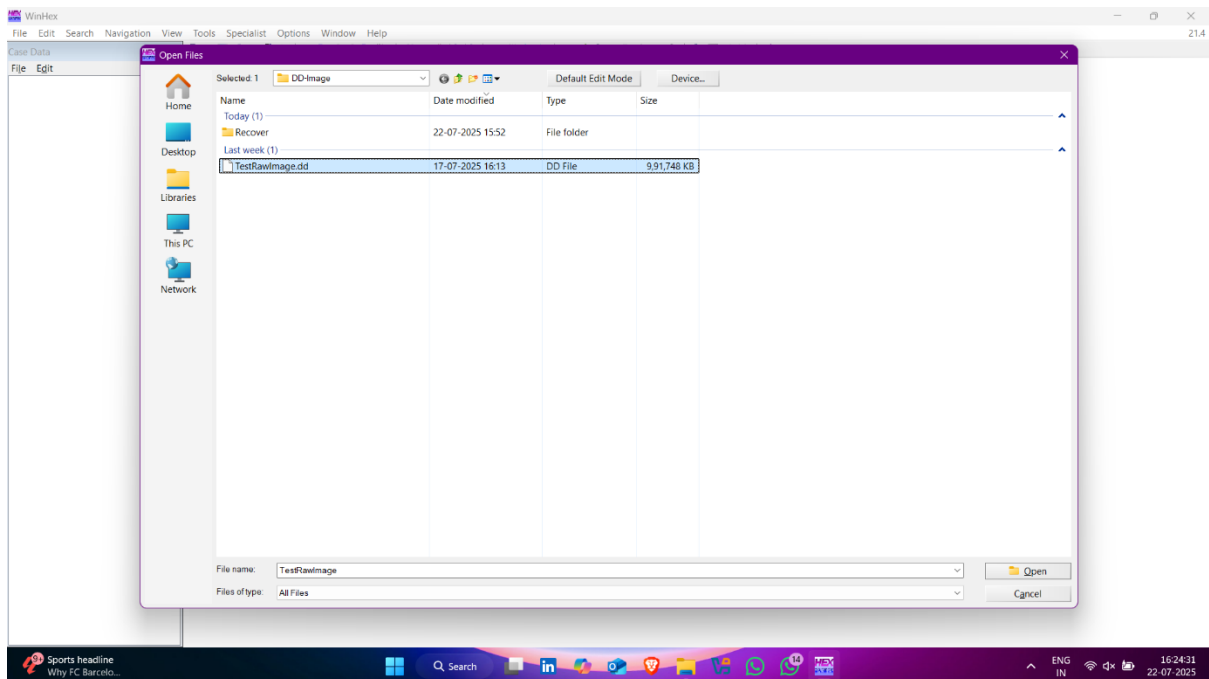
- Open Win Hex tool



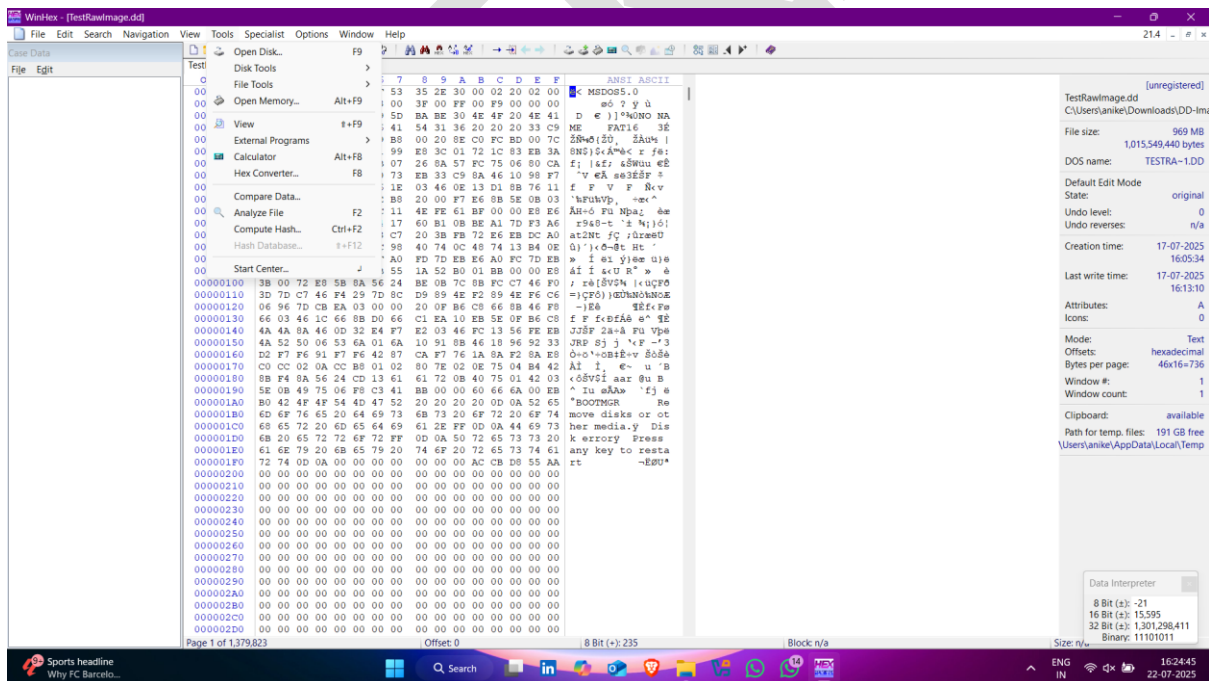
- Click on **File** Option and then click on **Open**



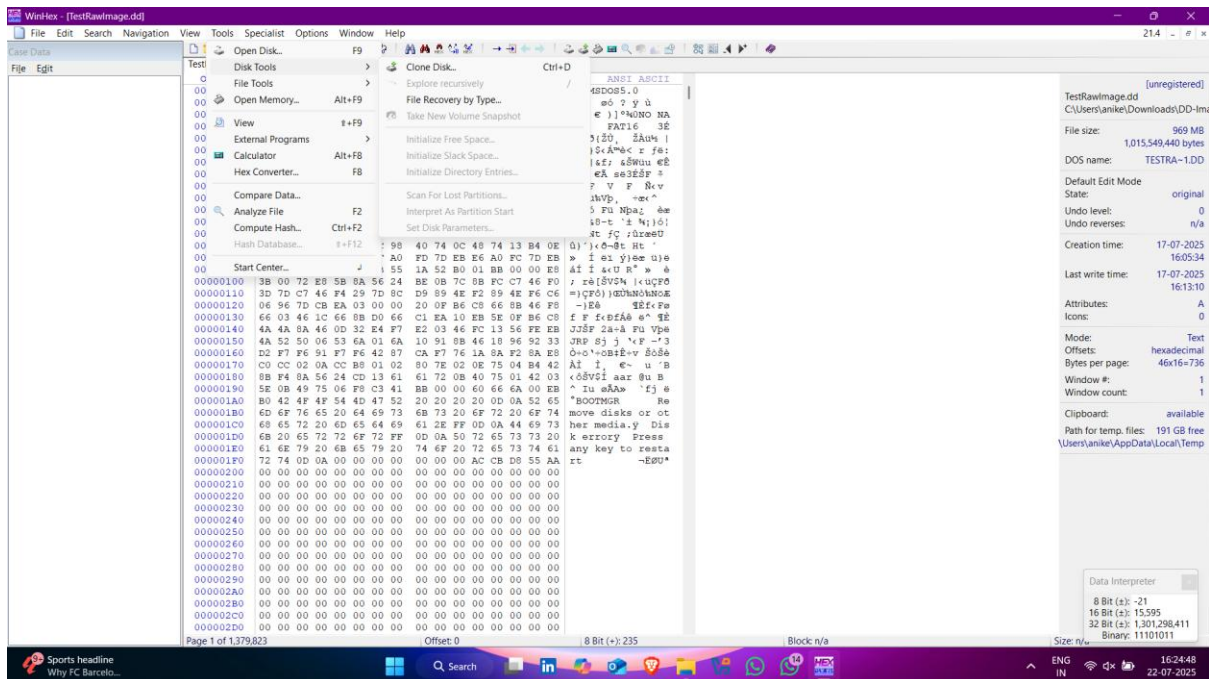
- Select **Image file** and click on **open**



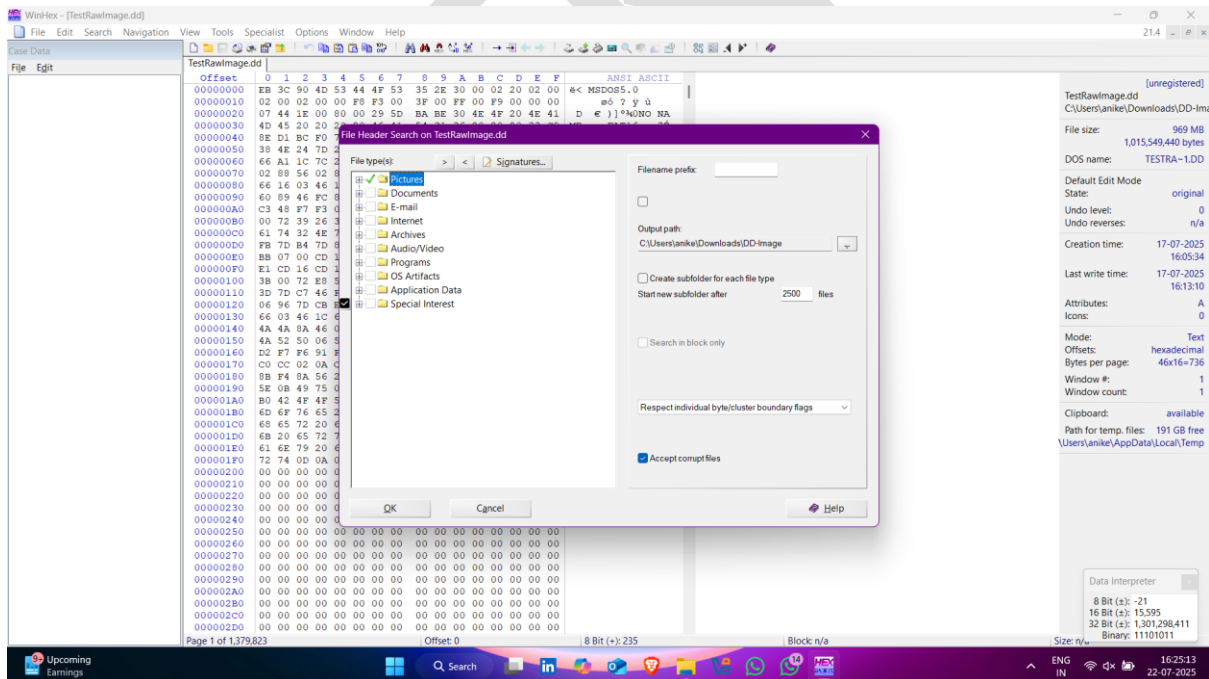
- Image open , now click on **Tools Section**



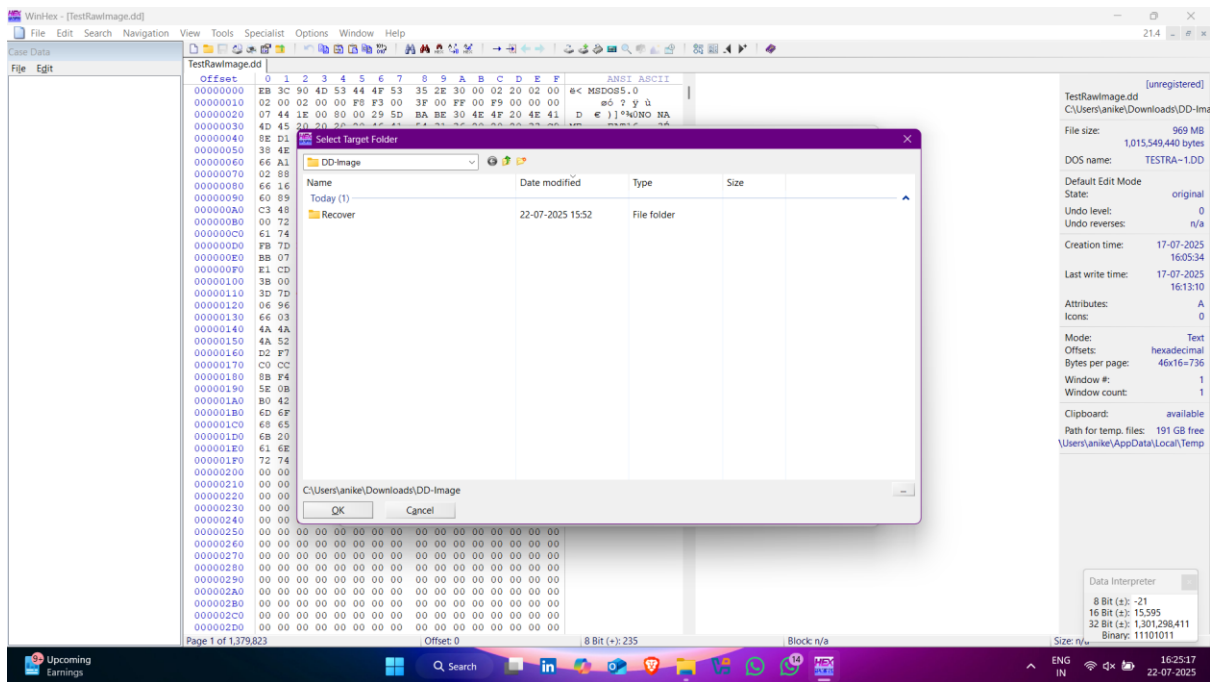
- Then click on Disk Tools And then file recovery type



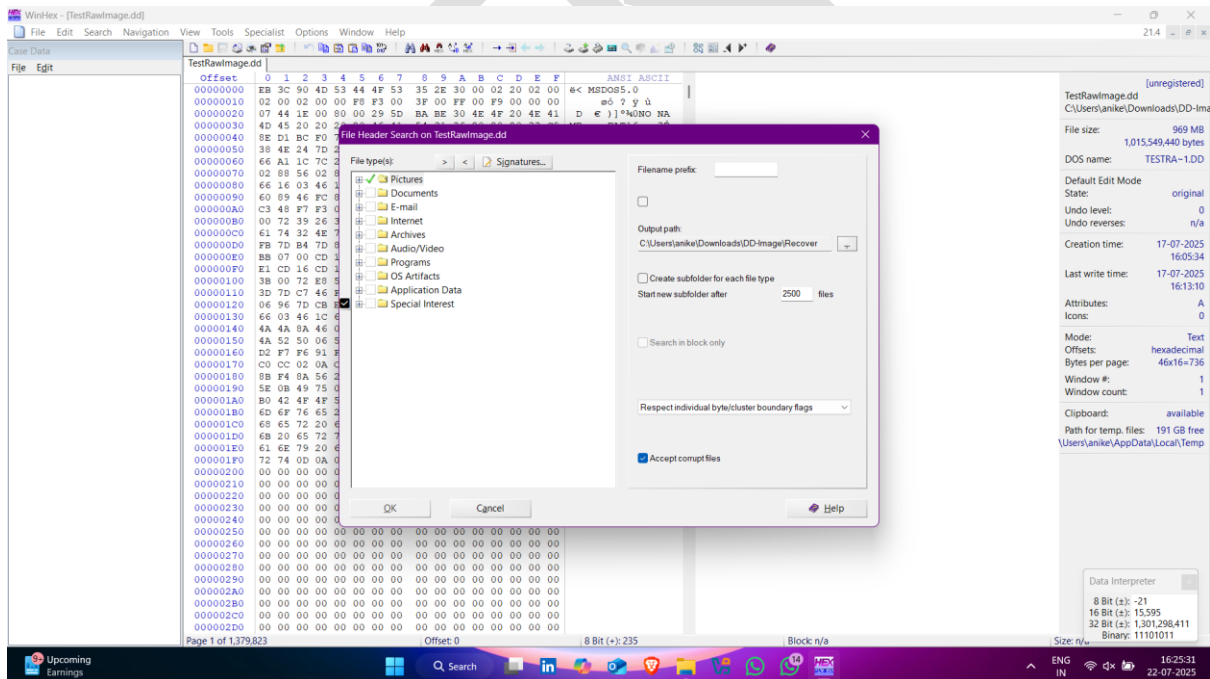
- Click on Down arrow (↓) to set destination for recover data



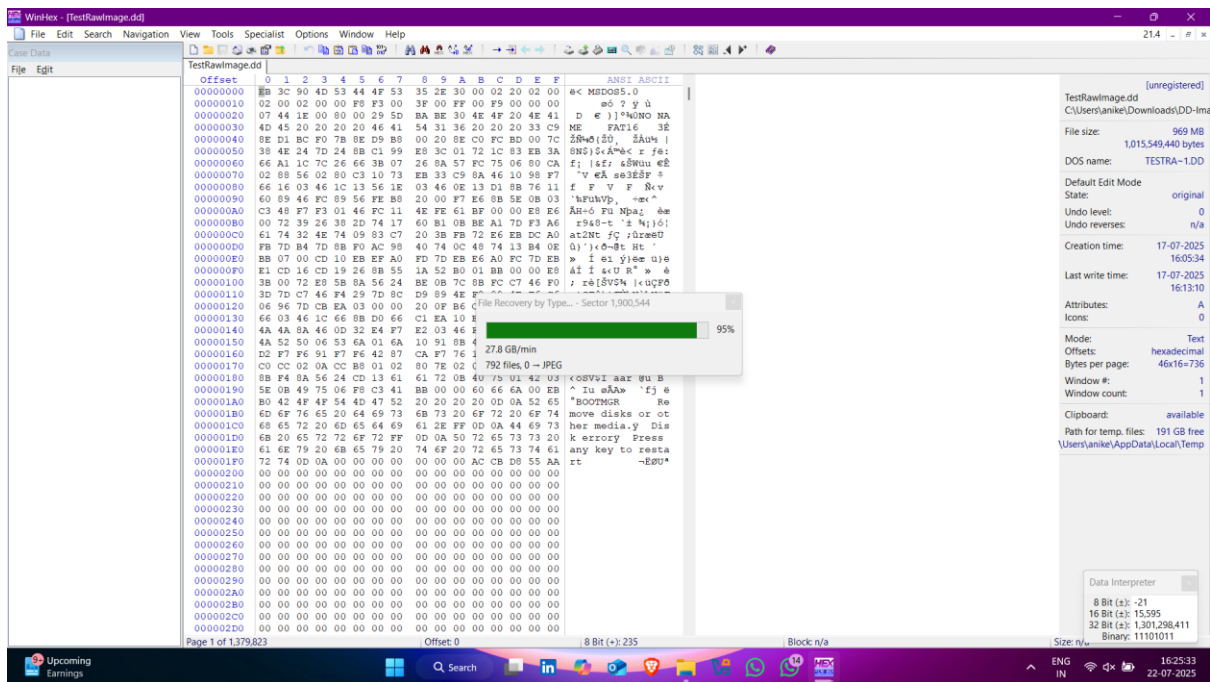
- Click on **Ok**



- Now click on **Ok**



- Data recovery Started  



- Data recover successfully  

